DIPLOMA PROGRAMME PRACTICE · SL · SET A

Chemistry / Paper 1A + 1B

Worked mark scheme · 55 marks · 90 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer B The dissolved salt passes through the filter. Iron and sand remain on it, where magnetic separation works after drying.

PAPER 1A · QUESTION 2 [1]

Answer D The 36-neutron isotope has mass number 65. With heavy fraction x, $63(1-x)+65x=64.30$, giving $x=0.650$.

PAPER 1A · QUESTION 3 [1]

Answer B X is Cl^- with 18 electrons. Removing its highest occupied 3p electron gives neutral chlorine.

PAPER 1A · QUESTION 4 [1]

Answer D The sample contains 0.0400 mol C and 0.0800 mol H. Their masses are 0.480 and 0.080 g. The remaining 0.320 g is 0.0200 mol O, giving 2:4:1.

PAPER 1A · QUESTION 5 [1]

Answer A Water still in the residue reduces apparent mass loss and is counted as dry salt. Both errors lower the calculated water-to-salt mole ratio.

PAPER 1A · QUESTION 6 [1]

Answer D $p_2=p_1(V_1/V_2)(T_2/T_1)=96.0\times1.50\times0.900=129.6$ kPa.

PAPER 1A · QUESTION 7 [1]

Answer A Two 3+ charges balance three 2− charges. In the melt the ions move, whereas in the solid they occupy fixed positions.

PAPER 1A · QUESTION 8 [1]

Answer D Vaporization separates intact HF molecules. Strong hydrogen bonding accounts for the unusually large intermolecular attraction.

PAPER 1A · QUESTION 9 [1]

Answer D NH_3 and H_2O each have four domains. One lone pair gives trigonal pyramidal NH_3 and two give bent H_2O.

PAPER 1A · QUESTION 10 [1]

Answer A Delocalized electrons carry current in both solids. Differently sized atoms distort the lattice and make slip harder.

PAPER 1A · QUESTION 11 [1]

Answer B Restore C=C between the two backbone atoms of one repeat to obtain $CH_2=CHCH_3$. Addition polymerization eliminates no small molecule.

PAPER 1A · QUESTION 12 [1]

Answer B For an isoelectronic series, the electron shells are comparable. Al^{3+} has 13 protons and attracts the electron cloud most strongly.

PAPER 1A · QUESTION 13 [1]

Answer C The sulfur oxidation numbers sum to +4 because oxygen contributes −6 and the overall charge is −2. Their average does not fix either individual value.

PAPER 1A · QUESTION 14 [1]

Answer C Both have formula C_3H_8O, but the first is an alcohol and the second an ether.

PAPER 1A · QUESTION 15 [1]

Answer B In butan-2-ol the OH-bearing carbon is attached to one methyl and one ethyl group. This is a secondary alcohol.

PAPER 1A · QUESTION 16 [1]

Answer A The solution gains $80.0\times4.18\times3.00=1003$ J. The reaction loses this heat, so $\Delta H=-1.003/0.0100\approx-100$ kJ mol^{-1}.

PAPER 1A · QUESTION 17 [1]

Answer A Reverse the second step to go from Y to Z. Its enthalpy becomes −17, giving $-42-17=-59$ kJ mol^{-1}.

PAPER 1A · QUESTION 18 [1]

Answer B Input=540 kJ, untransferred energy=216 kJ. Dividing by 18.0 g gives 12.0 kJ g^{-1}.

PAPER 1A · QUESTION 19 [1]

Answer B $C_3H_8+5O_2\rightarrow3CO_2+4H_2O$. Oxygen limits the reaction to 0.900/5=0.180 mol propane, producing 0.540 mol CO_2.

PAPER 1A · QUESTION 20 [1]

Answer C $n(CO_2)=0.0140$ mol equals $n(CaCO_3)$; its mass is 1.40 g. Purity=1.40/2.00×100.

PAPER 1A · QUESTION 21 [1]

Answer C $n(OH^-)=0.002160$ mol; n(acid)=0.001080 mol. Divide by 0.02000 dm^3.

PAPER 1A · QUESTION 22 [1]

Answer C A larger exposed surface increases collision frequency. Equal amounts of the limiting carbonate eventually yield equal amounts of gas.

PAPER 1A · QUESTION 23 [1]

Answer A Use both endpoints of the specified interval: $(50.0-22.0)/(55.0-15.0)=0.700$ cm^3 s^{-1}.

PAPER 1A · QUESTION 24 [1]

Answer B Compression favours the side with fewer gas molecules. The equilibrium constant depends on temperature.

PAPER 1A · QUESTION 25 [1]

Answer D $K_c=[HI]^2/([H_2][I_2])=0.36/0.020=18$.

PAPER 1A · QUESTION 26 [1]

Answer C $[H^+]=0.100\times10.0/40.0=0.0250$ mol dm^{-3}; $pH=-\log_{10}(0.0250)=1.60$.

PAPER 1A · QUESTION 27 [1]

Answer A Mg atoms oxidize to ions at the anode. The electrons pass along the wire to the copper cathode where Cu^{2+} is reduced.

PAPER 1A · QUESTION 28 [1]

Answer D $Mg^{2+}+2e^-\rightarrow Mg$ and $2Cl^-\rightarrow Cl_2+2e^-$ transfer the same two electrons per mole of product.

PAPER 1A · QUESTION 29 [1]

Answer C The first equation propagates the chain by replacing Cl• with an ethyl radical. Homolysis starts the chain and radical recombinations terminate it.

PAPER 1A · QUESTION 30 [1]

Answer A Hydroxide is the nucleophile. The C–Br bonding pair goes to bromine as substitution forms the C–O bond.

PAPER 1B · QUESTION 1 [12]

Gas collection from a mineral

(a) Write the net ionic equation and determine the mean rate of CO_2 production between 20 and 60 s. [3]

1 $CaCO_3(s)+2H^+(aq)\rightarrow Ca^{2+}(aq)+CO_2(g)+H_2O(l)$.

1 Use the difference in volume over this interval: (76−36)/(60−20).

1 1.00 $cm^3\ s^{-1}$.

(b) Calculate the mass percentage of calcium carbonate in the sample. [3]

1 96 cm^3=0.096 dm^3; amount CO_2=0.096/24.0=0.00400 mol.

1 The 1:1 ratio gives 0.00400×100.1=0.4004 g $CaCO_3$.

1 100×0.4004/0.500=80.1% (accept 80%).

(c) The final collected volume has absolute uncertainty ±1 cm^3 and the sample mass ±0.002 g. Treat other data as exact. Estimate the maximum absolute uncertainty in the percentage purity. [3]

1 Relative volume uncertainty is 1/96=1.04%.

1 Relative mass uncertainty is 0.002/0.500=0.40%; add for a quotient to obtain 1.44%.

1 80.1×0.0144≈1.2 percentage points. Accept appropriate rounding.

(d) A repeat with smaller particles reaches the same plateau sooner. Explain both observations and predict the effect of an unnoticed gas leak on the calculated purity. [3]

1 Smaller particles expose greater surface area to the acid, increasing collision frequency and rate.

1 Equal initial amounts of carbonate produce the same final CO_2 at the same temperature and pressure.

1 A leak reduces collected CO_2 so the inferred carbonate purity is too low.

PAPER 1B · QUESTION 2 [13]

Recovering the mixing temperature

(a) Use the cooling data to estimate the temperature immediately after mixing and the corrected temperature rise. [3]

1 Gradient is −0.20 °C min^{-1}.

1 Extrapolation back one minute from 28.6 °C gives 28.8 °C.

1 Corrected rise is 28.8−21.0=7.8 K.

(b) Calculate the energy gained by the solution and by the cup, separately. State which quantity must be used as the magnitude of the heat released. [3]

1 Solution mass=100 g, so qsolution=100×4.18×7.8=3.26 kJ.

1 qcup=36.0×7.8=0.281 kJ.

1 The reaction supplied both, a total of 3.54 kJ.

(c) Write the net ionic reaction and calculate ΔH per mole of water formed. [3]

1 $H^+(aq)+OH^-(aq)\rightarrow H_2O(l)$.

1 Each reactant supplies 0.0500×1.20=0.0600 mol; 0.0600 mol H_2O forms.

1 $\Delta H=-3.5412/0.0600\approx-59.0$ kJ mol^{-1}; accept −59.

(d) Explain the direction of bias if the first post-mixing temperature were used without extrapolation and the cup heat capacity were also ignored. [2]

1 Using 28.6 °C instead of 28.8 °C understates the rise and heat released.

1 Ignoring heat stored in the cup further underestimates the magnitude, giving a less negative enthalpy.

(e) Give one condition needed for this extrapolation and one experimental check of that condition. [2]

1 Cooling must be approximately linear over the short interval used, with rapid mixing/reaction relative to the observation period.

1 Collect additional closely spaced post-mixing readings and check residuals or graphical linearity; accept a suitable repeat at higher sampling frequency.